Customer Churn Data Cleaning Report

Source: customer_churn_dataset-testing-master.csv

Generated: 2026-08-28 15:35

Input: 64,374 rows x 12 columns

Output: 64,374 rows x 12 columns

Purpose: make the dataset consistent, valid, and ready for analysis while capping extreme numeric values at documented percentile limits.

Overall result: 1,260 extreme numeric values were capped; 0 rows were removed by quality checks.

Before and After Quality Metrics

Metric	Before cleaning	After cleaning
Rows	64374	64374
Columns	12	12
Missing values	0	0
Blank strings	0	0
Exact duplicate rows	0	0
Unique CustomerID	64374	64374
Sequential IDs	True	True

Cleaning Rules and Audit Results

Check	Rule	Status	Affected rows
Column names	Trim leading/trailing whitespace	PASS	0
Text fields	Trim leading/trailing whitespace	PASS	0
Numeric fields	Convert to numeric; invalid values b	PASS	0
Exact duplicate rows	Remove duplicate records	PASS	0
Required ranges	Validate domain ranges	PASS	0
Allowed categories	Validate categorical domains	PASS	0
Numeric outliers	Cap values at the 1st and 99th perce	PASS	1260
Rows retained	Keep records passing all checks	INFO	64374
CustomerID	1 to inf	PASS	0
Age	18 to 100	PASS	0
Tenure	0 to inf	PASS	0
Usage Frequency	0 to inf	PASS	0
Support Calls	0 to inf	PASS	0
Payment Delay	0 to inf	PASS	0
Total Spend	0 to inf	PASS	0
Last Interaction	0 to inf	PASS	0
Churn	0 to 1	PASS	0
Gender	Female, Male	PASS	0
Subscription Type	Basic, Premium, Standard	PASS	0
Contract Length	Annual, Monthly, Quarterly	PASS	0
Age	Cap to 18.00 through 65.00	PASS	0
Tenure	Cap to 1.00 through 60.00	PASS	0
Usage Frequency	Cap to 1.00 through 30.00	PASS	0
Support Calls	Cap to 0.00 through 10.00	PASS	0
Payment Delay	Cap to 0.00 through 30.00	PASS	0
Total Spend	Cap to 109.00 through 990.00	PASS	1260
Last Interaction	Cap to 1.00 through 30.00	PASS	0

Cleaned Dataset Column Profile

Column	Data type	Non-null	Unique	Minimum	Maximum
CustomerID	int64	64374	64374	1	64374
Age	int64	64374	48	18	65
Gender	string	64374	2	-	-
Tenure	int64	64374	60	1	60
Usage Frequency	int64	64374	30	1	30
Support Calls	int64	64374	11	0	10
Payment Delay	int64	64374	31	0	30
Subscription Type	string	64374	3	-	-
Contract Length	string	64374	3	-	-
Total Spend	int64	64374	882	109	990
Last Interaction	int64	64374	30	1	30
Churn	int64	64374	2	0	1

Numeric Validation Summary

Column	count	mean	std	min	25%	50%	75%	max
CustomerID	64374.0	32187.5	18583.32	1.0	16094.25	32187.5	48280.75	64374.0
Age	64374.0	41.97	13.92	18.0	30.0	42.0	54.0	65.0
Tenure	64374.0	31.99	17.1	1.0	18.0	33.0	47.0	60.0
Usage Frequenc	64374.0	15.08	8.82	1.0	7.0	15.0	23.0	30.0
Support Calls	64374.0	5.4	3.11	0.0	3.0	6.0	8.0	10.0
Payment Delay	64374.0	17.13	8.85	0.0	10.0	19.0	25.0	30.0
Total Spend	64374.0	541.02	260.7	109.0	313.0	534.0	768.0	990.0
Last Interaction	64374.0	15.5	8.64	1.0	8.0	15.0	23.0	30.0
Churn	64374.0	0.47	0.5	0.0	0.0	0.0	1.0	1.0

Key Data-Cleaning Details

1. Ingestion: read the CSV with pandas and verify that the expected customer-churn schema is present.

2. Column names: remove accidental leading or trailing whitespace so downstream references are stable.

3. Text standardization: trim whitespace in text fields; preserve the meaningful category labels.

4. Type validation: coerce expected numeric fields to numeric values and flag conversion failures as missing.

5. Missingness: measure null values and blank strings separately. Missing values should be imputed only with a documented business rule; none are present here.

6. Duplicates: check full-row duplicates and CustomerID uniqueness. Duplicate entities should be investigated before aggregation or modeling.

7. Range checks: verify Age is at least 18, operational measures are non-negative, and Churn is binary 0/1.

8. Category checks: verify Gender, Subscription Type, and Contract Length against the observed domain definitions.

9. Outliers: cap configured numeric fields at their 1st and 99th percentiles; this limits extreme influence without deleting valid customer records.

10. Reproducibility: the cleaned file and row-level audit summary are written to the output folder.

Outlier Analysis and Treatment

Definition: an outlier is an observation unusually far from most other values. It may be a data-entry error, a system error, or a genuine customer behavior pattern.

Detection method 1 - IQR: calculate IQR = Q3 - Q1. Flag values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR.

Detection method 2 - Percentiles: flag values below the 1st percentile or above the 99th percentile. This project uses percentile limits for capping.

Detection method 3 - Z-score: calculate z = (value - mean) / standard deviation. Values with an absolute z-score above 3 may be potential outliers when data is approximately normally distributed.

Mean: uses every value but is strongly affected by extreme values. Use it for roughly symmetric data with few outliers, or for comparison during analysis.

Median: the middle sorted value and resistant to extreme values. Prefer it for skewed numeric fields such as Total Spend and for numeric missing-value imputation.

Mode: the most frequent value. Use it mainly to fill missing categorical fields such as Gender, Subscription Type, and Contract Length; it is not normally used to treat numeric outliers.

Removal: delete a record only when the value is impossible or clearly incorrect. Do not delete a valid high-value customer simply because the value is unusual.

Capping: replace values below the lower limit with the lower limit and values above the upper limit with the upper limit. This project caps selected numeric fields at the 1st and 99th percentiles.

Transformation: a log transformation such as log1p(Total Spend) can reduce right skew while retaining the customer record.

Project treatment: Age, Tenure, Usage Frequency, Support Calls, Payment Delay, Total Spend, and Last Interaction are checked and capped. CustomerID and Churn are not capped.

Observed result: 1,260 values were capped, and 0 rows were removed by the other quality checks.

Best practice: investigate unusual values, document the chosen treatment, apply the same rule to future data, and fit model preprocessing using training data only.

Recommendations and Handoff

The current file is structurally clean and no records were removed.

Keep the audit CSV with the cleaned extract so every transformation is reviewable.

For future files, stop or quarantine rows that fail range/category checks rather than silently dropping them.

Confirm business definitions for Age, Total Spend, Payment Delay, and Tenure before modeling.

Encode categorical fields and scale numeric features only in the modeling pipeline, fitted on training data to prevent leakage.

Re-run this script whenever the source CSV changes; outputs are deterministic apart from the report timestamp.